Virtualization

Objectives:

- Upon completion of this section, you will:
  - Be able to describe what virtualization is.
  - Understand virtualization types.
  - Understand type-2 technology.
  - Understand type-1 technology.

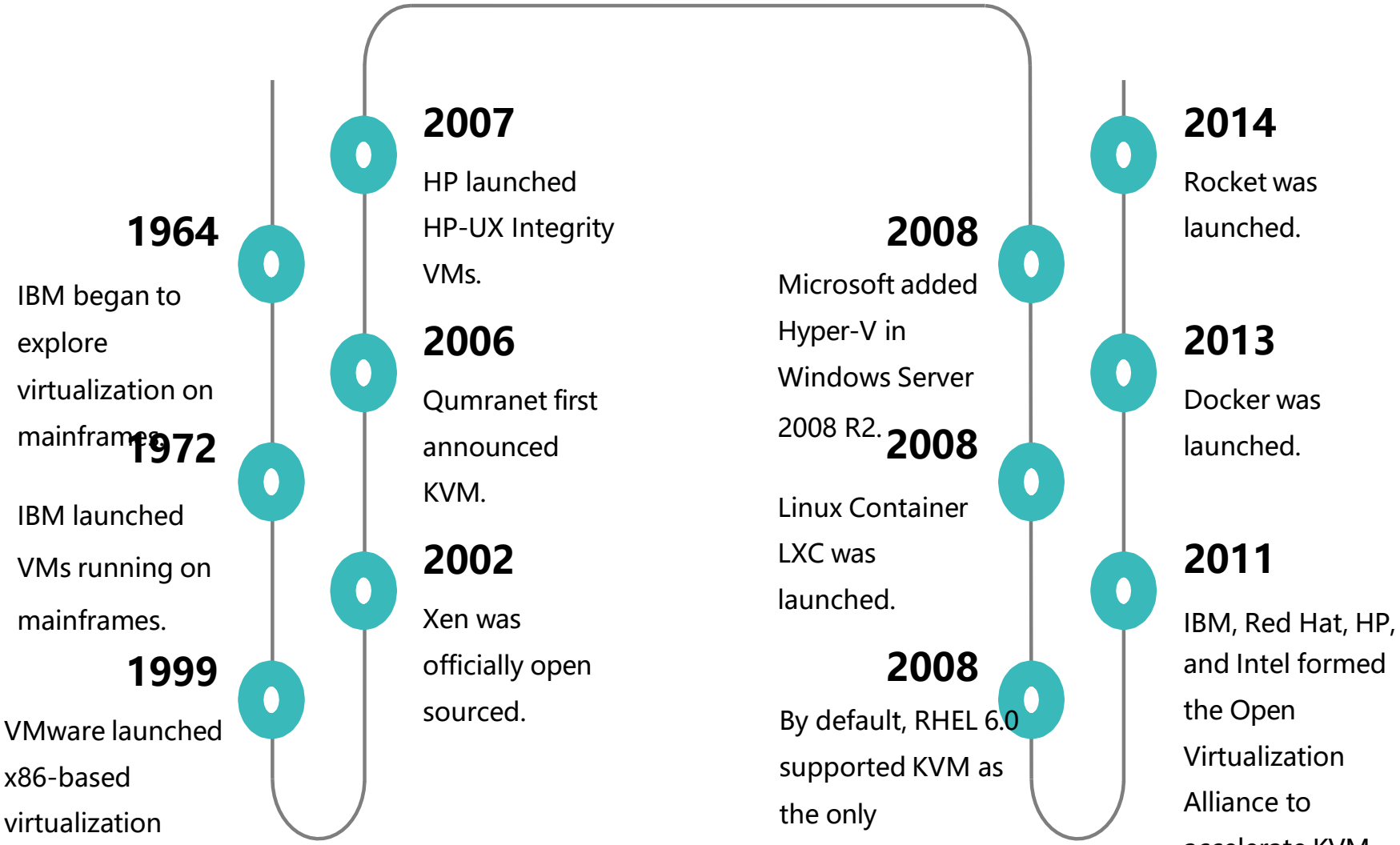
Contents:

- History of Compute Virtualization.
- Important Concepts in Compute Virtualization.
- Types of Compute Virtualization.
- Introduction to hypervisor type-2.
- Hypervisor type-2 examples.
- VMware workstation as an example.
- LAB:
  - Install VMware workstation.
  - Deploy, manage and access VMs.

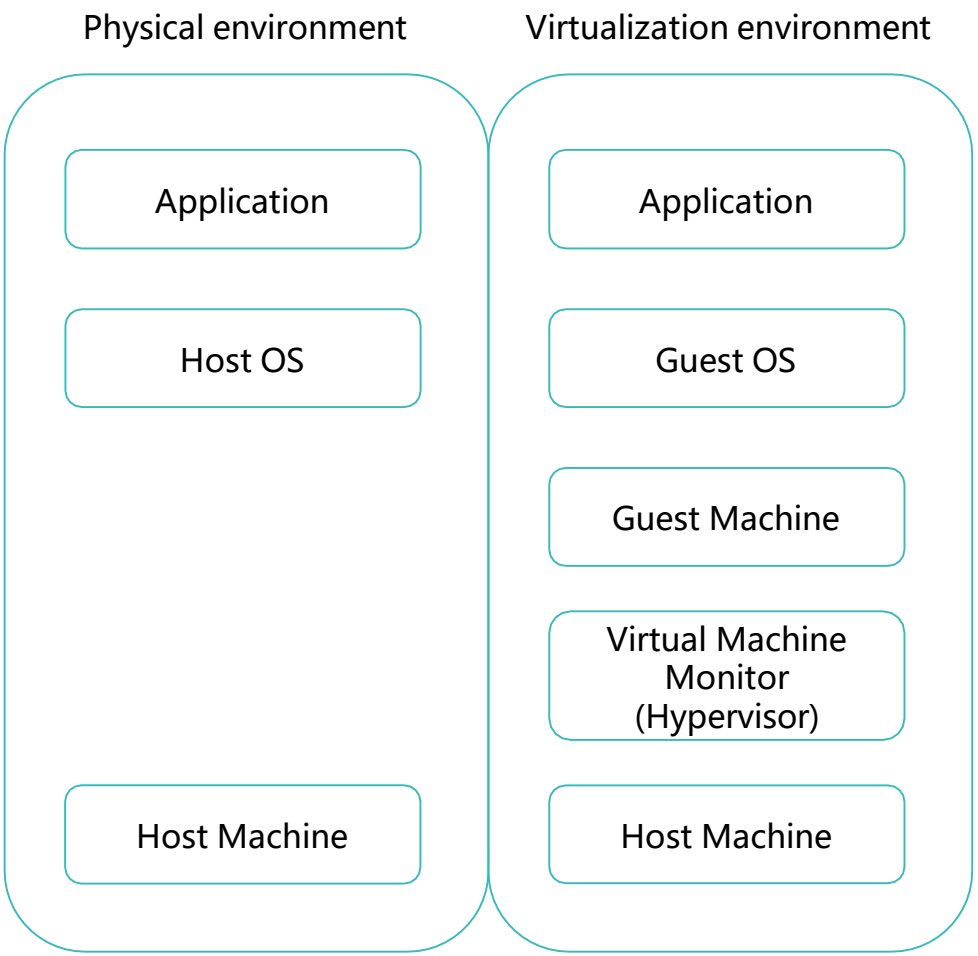
Contents:

- Introduction to hypervisor type-1.
- Hypervisor type-1 examples.
- ESXI as an example.
- LAB:
  - Install ESXI and vCenter.
  - Deploy, manage and access VMs.

History of Compute Virtualization:



Important Concepts in Compute Virtualization:



Guest OS:
Operating system running in a virtual machine (VM)

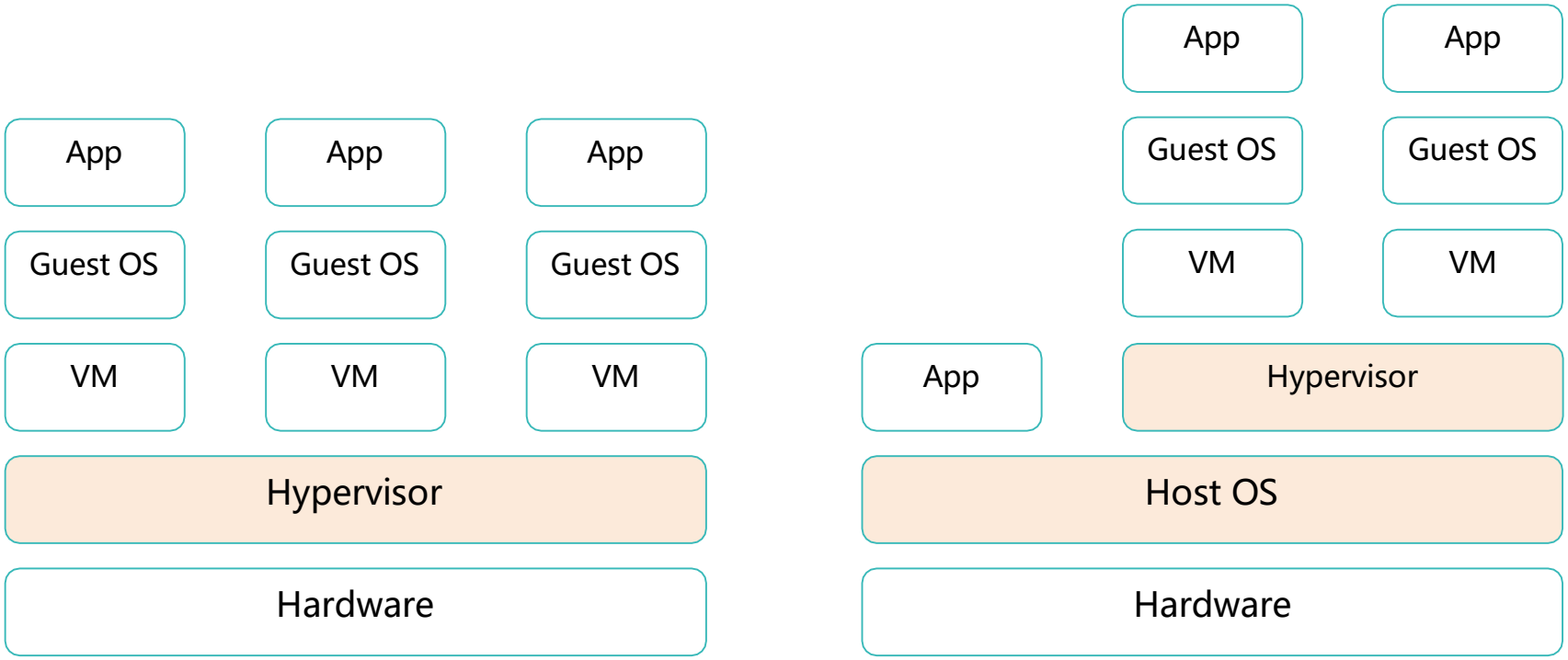
Guest Machine:
Virtual machine created through virtualization

Hypervisor:
Virtualization software layer, or Virtual Machine Monitor (VMM)

Host OS:
Operating system running in a physical machine

Host Machine:
Physical machine

Types of Compute Virtualization:

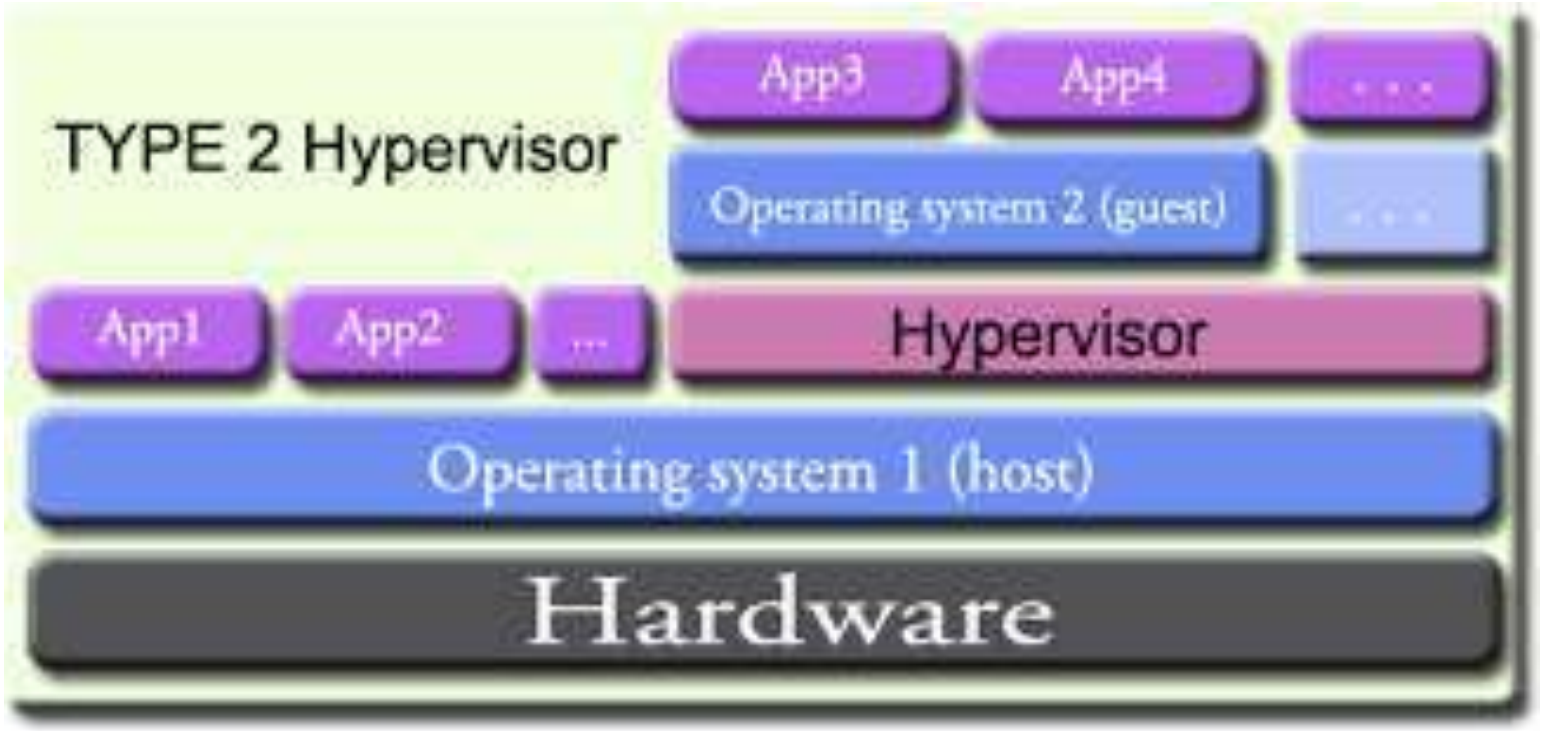


Bare-Metal Virtualization
(Type 1)

Hosted Virtualization
(Type 2)

Hypervisor type-2:

- Hypervisors run on a host operating system that provides virtualization services, such as I/O device support and memory management.



Hypervisor type-2 examples:

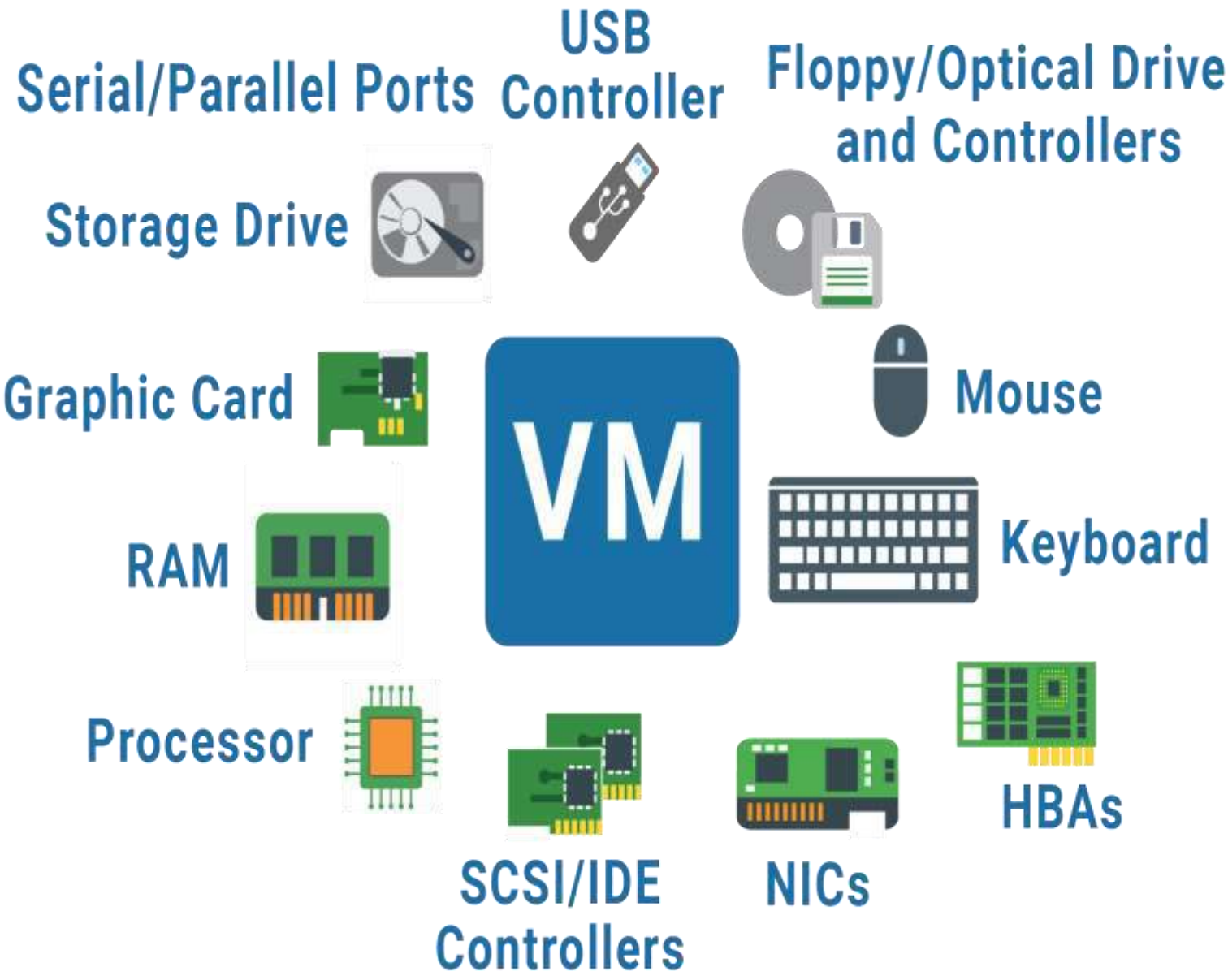
- VMware Workstation/Fusion/Player.
- Microsoft Virtual PC.
- Oracle VM VirtualBox.
- Red Hat Enterprise Virtualization.

VMware workstation:

- VMware Player is a free virtualization hypervisor.
- VMware Workstation is a more robust hypervisor with some advanced features, such as record-and-replay and VM snapshot support.
- VMware Workstation has three major use cases:
  - for running multiple different operating systems or versions of one OS on one desktop.
  - for developers that need sandbox environments and snapshots.
  - for labs and demonstration purposes.

WHAT IS A VM?

- VM stands for “Virtual Machine”
- Virtualization creates virtual hardware by cloning physical hardware
- The hypervisor uses virtual hardware to create a virtual machine (VM)
- A VM is a set of files
- With a hypervisor running VMs, one computer can run multiple OS simultaneously
- A VM can run applications like a physical machine

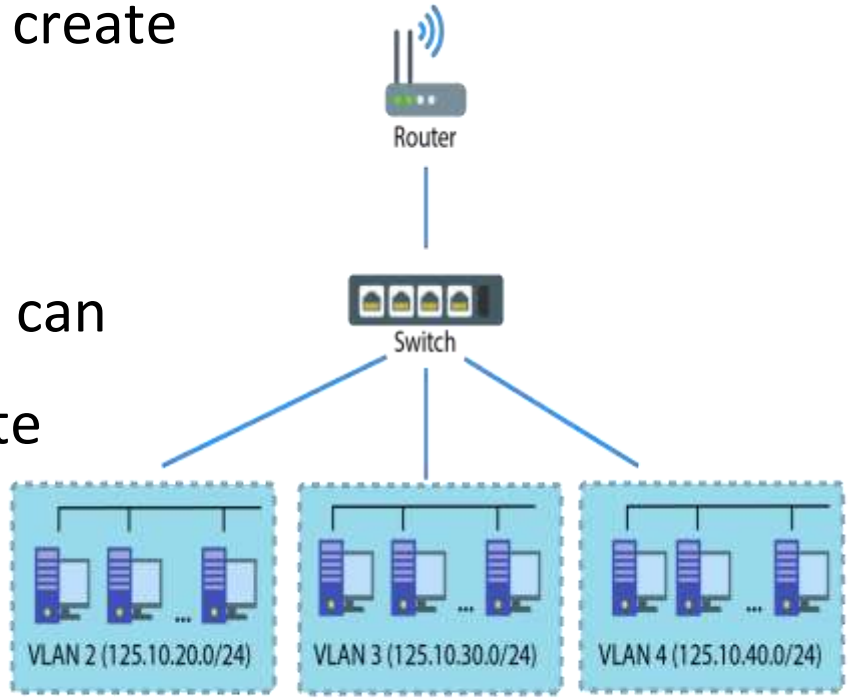

Serial/Parallel Ports
USB Controller
Floppy/Optical Drives and Controllers
Storage Drive
Graphic Card
VM
Mouse
RAM
Keyboard
Processor
SCSI/IDE Controllers
NICs
HBAs

NETWORK VIRTUALIZATION

- Physical components that make up the physical network are virtualized to create a virtual network
- What is a vSwitch?

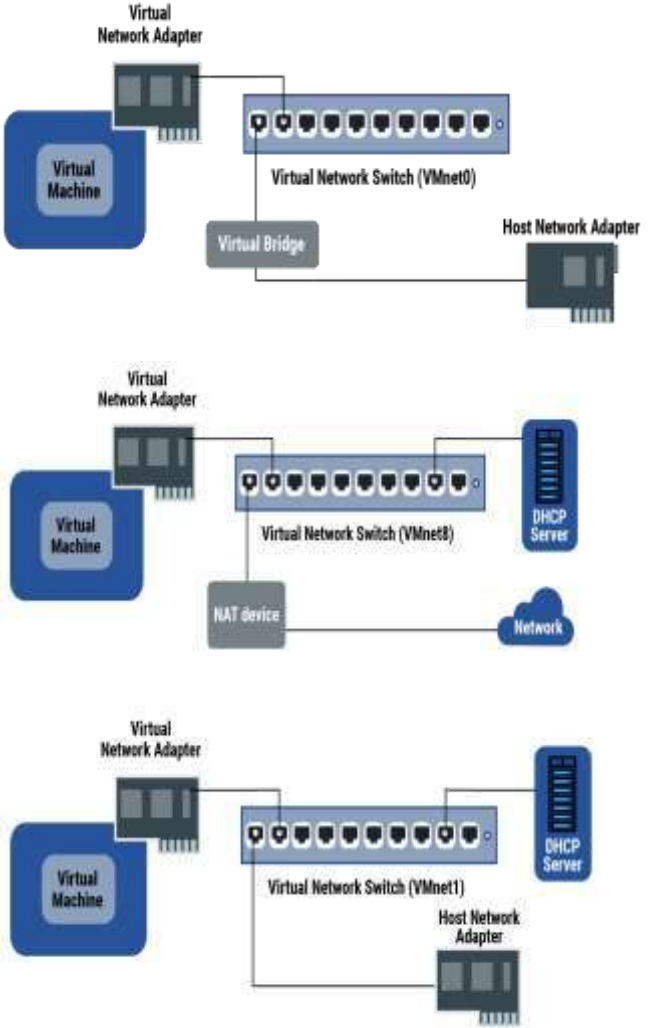
  - Virtual switch that virtual devices can connect to in order to communicate with each other
- What is a vLAN?

  Virtual Local Area Network that is segmented into groups of ports isolated from one another, creating different network segments



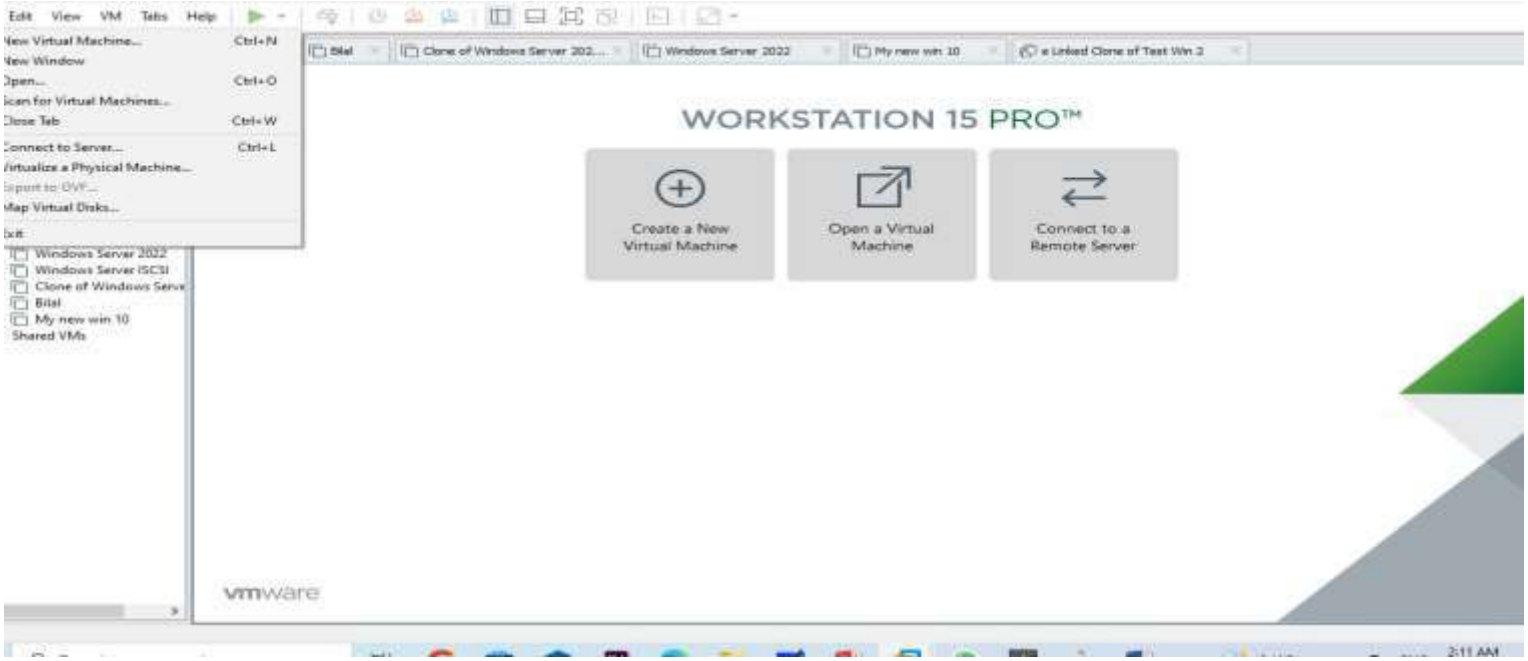
TYPES OF VIRTUAL NETWORKS

- **Bridged Network**: The host server and the VM are connected to the same network, and the host shares its IP address with the VM
- **NAT**: VMs use an IP translated from the host's IP (using NAT device) and communicate on a private network .
- **Host-only Network**: VMs use a private network but do not have translated IP addresses to connect to an external network.



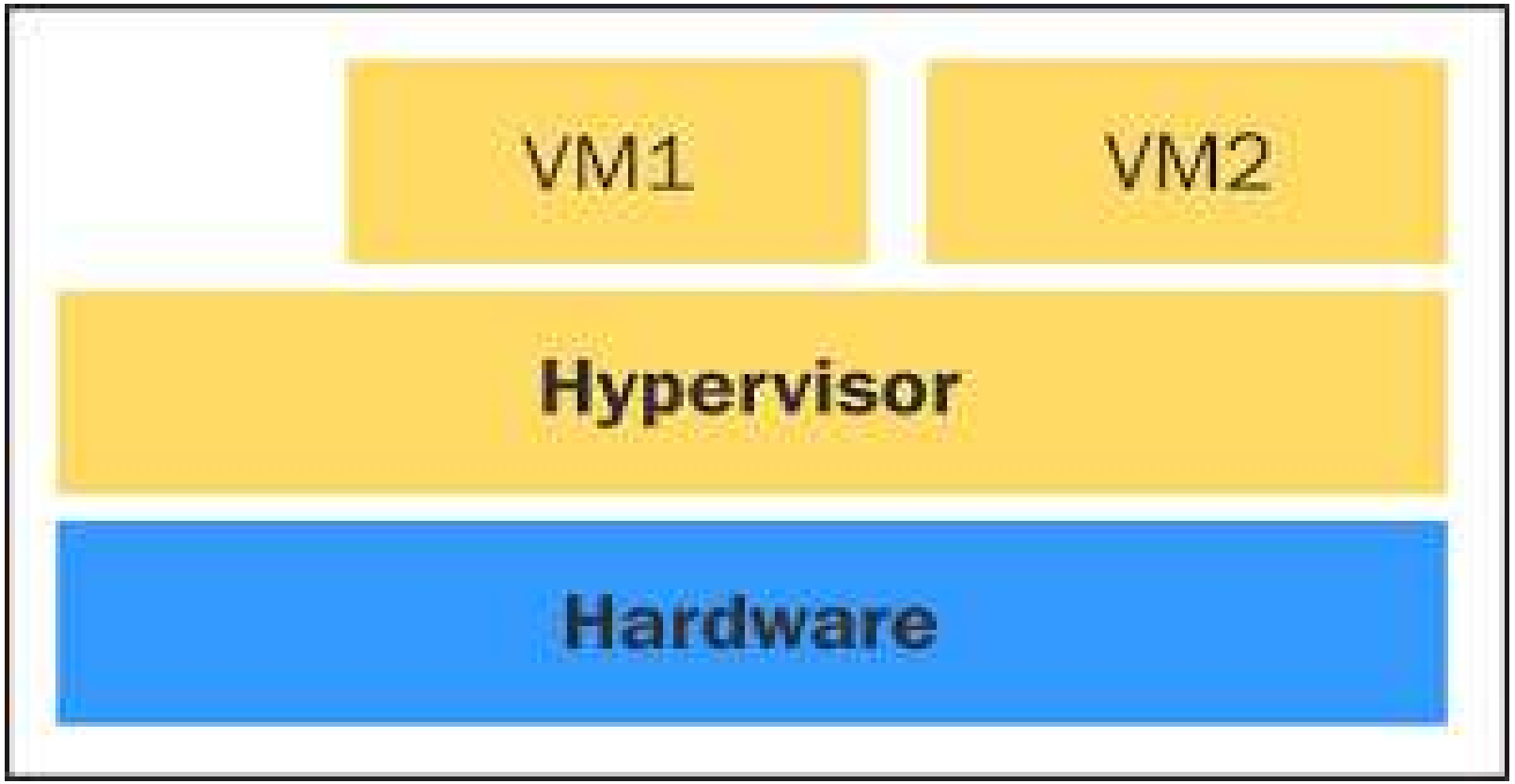
LAB:

- Install VMware workstation.
- Deploy, manage VMS.
- Access VMs remotely.



Hypervisor type-1:

- Hypervisors run directly on the system hardware – A “bare metal” embedded hypervisor.



Hypervisor type-1 examples:

- VMware ESX and ESXi.
- Microsoft Hyper-V.
- Citrix Xenserver.
- Oracle VM.

Citrix Xenserver:

- Is an open source server virtualization platform based on the Xen hypervisor.
- Citrix also offers a supported version that you can purchase, with two options: Standard and Enterprise.
- This platform is used by virtualization administrators to deploy, host and manage VMs.
- It's also used to distribute hardware resources -- CPU, memory, networking, storage -- to VMs.

VMWare ESXI:

- a bare-metal Type-1 hypervisor developed by Broadcom (via VMware).
- installs directly onto physical server hardware without needing a general-purpose operating system.
- This platform is used to deploy, host and manage VMs.
- ESXI has Lightweight Footprint
- It's also used to distribute hardware resources -- CPU, memory, networking, storage -- to VMs.

LAB:

- Install ESXI and vCenter.
- Deploy, manage VMS.
- Access VMs remotely using RDP or SSH.